

Aggregating Judgments when People Receive Advice from an Artificial Intelligence System

Preregistration

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Overview

In many managerial settings, good decision making depends on obtaining an accurate forecast of an uncertain variable of interest. Useful knowledge about the variable may be accessible to both human experts and AI models, where each may have their own relative advantages in forming an accurate forecast. To make use of both sources of information, AI advice can be provided to human experts who can assess how much weight it should be given when updating their prior forecasts. Since useful information may be held by different individuals, combining forecasts from multiple experts can also boost the accuracy of the aggregate forecast. However, while providing AI advice to humans can be helpful at an individual level, it may also induce correlations in judgment errors that hamper efficient combination of information at the aggregate level. Using a stylized Bayesian model of information held by human experts and the algorithm, we have derived a new procedure for aggregating judgments when humans receive AI advice. The method uses individual responses to estimate how much weight to put on human versus AI forecasts and forms an aggregate forecast by taking the final average human forecast and adjusting it back toward the initial average human forecast.

The purpose of this online experiment is to evaluate the performance of this proposed judgment aggregation procedure and to test whether it can outperform existing alternatives.

Our primary research questions are:

- (1) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average initial human forecast?
- (2) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the average final human forecast?
- (3) Are the aggregate forecasts produced from this proposed judgment aggregation procedure more accurate than the AI advice?

Details of our proposed judgment aggregation method:

In our setting, a crowd of J judges each begin by providing their own, individual estimate of the variable of interest, before receiving any advice. The true value of the unknown variable of interest is θ . We denote the initial estimate given by judge j by f_j^0 . Next, all judges are provided with the same algorithmic advice, denoted by f^a , and asked to update their estimate of the variable of interest. We denote the final estimate given by judge j by f_j^1 . We denote the average of the initial and final estimates over all J judges as $\bar{f}^0$ and $\bar{f}^1$, respectively.

Once this regression has been estimated, we form our proposed aggregate forecast as follows:

$$\hat{\theta} = \bar{f}^1 + \gamma_R(A)(\bar{f}^0 - \bar{f}^1),$$

$\gamma_R(A)$ is defined below:

DEFINITION 10. $\gamma_R(A)$ is obtained by solving the following nested adversarial optimization problem, conditioning on the DM's prior belief $\mathcal{A}$:

$$\left(\hat{\rho}_{h,a}(\tilde{\gamma}), \hat{\rho}_{h,h}(\tilde{\gamma}), \hat{\phi}_0(\tilde{\gamma}), \hat{\beta}(\tilde{\gamma}) \right) \in \hat{\mathcal{A}}(\tilde{\gamma}) = \underset{(\rho_{h,a}, \rho_{h,h}, \phi_0, \beta) \in \mathcal{A}'}{\operatorname{argmax}} \mathcal{R}(\tilde{\gamma} | \rho_{h,a}, \rho_{h,h}, \phi_0, \beta, J); \quad (\text{IP})$$

$$\gamma_R(A) \in \Gamma_R(A) = \underset{\tilde{\gamma} \in \mathbb{R}}{\operatorname{argmin}} \mathcal{R}(\tilde{\gamma} | \hat{\rho}_{h,a}(\tilde{\gamma}), \hat{\rho}_{h,h}(\tilde{\gamma}), \hat{\phi}_0(\tilde{\gamma}), \hat{\beta}(\tilde{\gamma}), J), \quad (\text{OP})$$

where

$$\mathcal{A}' = \mathcal{A} \cap \left\{ (\phi_0, \rho_{h,a}, \rho_{h,h}, \beta) : \phi_0 > \rho_{h,a}, \rho_{h,h} > \phi_0 \rho_{h,a}; \beta \in \left\{ \underline{\beta}(\rho_{h,a}, \rho_{h,h}, \phi_0), \bar{\beta}(\rho_{h,a}, \rho_{h,h}, \phi_0) \right\} \right\}. \quad (20)$$

The regret function used to define $\gamma_R(A)$ is described below:

(e) *The normalized parametric regret $\mathcal{R}(\gamma)$ expressed as a function of the three forecast parameters is as follows:*

$$\begin{aligned} \mathcal{R}(\gamma) &\equiv \mathcal{R}(\gamma | \rho_{h,a}, \rho_{h,h}, \phi_0, \beta, J) \\ &= \left(\beta(1-\gamma) \frac{\frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}}}{\frac{\phi_0}{\rho_{h,a}} - 1 + \frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}}} \right)^2 \cdot \left(\frac{\phi_0}{\rho_{h,a}} - 1 \right) \\ &\quad + \left(1 - \beta(1-\gamma) \frac{\frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}}}{\frac{\phi_0}{\rho_{h,a}} - 1 + \frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}}} \right)^2 \cdot \left(\frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1}{J} \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}} \right) \\ &\quad - \frac{\left(\frac{\phi_0}{\rho_{h,a}} - 1 \right) \left(\frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 + \frac{1}{J} \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}} \right)}{\left(\frac{\phi_0}{\rho_{h,a}} - 1 \right) + \left(\frac{\rho_{h,h}}{\phi_0 \rho_{h,a}} - 1 \right) + \frac{1}{J} \frac{1-\rho_{h,h}}{\phi_0 \rho_{h,a}}}. \end{aligned} \quad (18)$$

To obtain the set A for each question, we will use the methods described as follows. We assume each of the tasks to be perfectly replaceable, and will employ a holdout approach with random sampling of individual forecasts from each task. To estimate ranges for each of ρ_{ha} , ϕ_0 , and β for task i , we will select one response at random from each question the holdout set of $N=199$, where each task is indexed by k . Let f_k^0 and f_k^a be the randomly sampled initial human forecast and the AI advice respectively for the k -th holdout task, and let θ_k be the true value of the task. We will compute the following estimates from the randomly sampled data shown below.

$$\hat{\rho}_{h,a} = \text{Cor}(f_k^0 - \theta_k, f_k^a - \theta_k); \quad (21)$$

$$\hat{\phi}_0 = \sqrt{\frac{\frac{1}{N} \sum_{k=1}^N (f_k^a - \theta_k)^2}{\frac{1}{N} \sum_{k=1}^N (f_k^0 - \theta_k)^2}}. \quad (22)$$

To estimate ranges for β , we will first calculate $\hat{w}_k$ for every task k using the regression $f_{jk}^1 - f_{jk}^0 = w_k(f_k^a - f_{jk}^0) + \xi_{jk}$ which uses *all recorded responses* for task k . $\hat{w}_k$ will be the empirically observed average WOA used by the group of forecasters for task k . Then we will estimate $\hat{\beta}$ as the algorithm aversion factor for which the mean squared error of a counterfactual crowd containing N rational, un-averse forecasters predicting N different questions would be minimized. The minimization problem is shown below.

$$\hat{\beta} \in \hat{B} = \underset{\beta > 0}{\text{argmin}} \frac{1}{N} \sum_{k=1}^N \left(\left(\frac{\hat{w}_k}{\beta} f_k^a + \left(1 - \frac{\hat{w}_k}{\beta} \right) f_k^0 \right) - \theta_k \right)^2. \quad (23)$$

Finally, for each of the three parameters, we will independently repeat the random sampling and estimation process 5000 times to create an empirical distribution of the parameter estimates, and generate a 95% confidence interval for the parameters.

Unlike the earlier parameters, estimating ρ_{hh} will require observing variation in errors within multiple human forecasts for each task. To do this, we will calculate the intra-class correlation coefficient (ICC) using a linear mixed-effects regression as the proportion of the total variation in initial forecast errors that can be explained by task fixed-effects. We will perform the regression given below across all recorded responses for N tasks, and then use the *performance* package in R to calculate a 95% confidence interval of the ICC obtained from the regression equation.

$$f_{jk}^0 - \theta_k = \mu + v_k + v_{jk} \quad (24)$$

R code for the estimation steps is provided below:

```

n_reps <- 5000
tasks <- 200
sample_correlation_ha <- function(df) {
  sampled_rows <- df %>% group_by(Task) %>% slice_sample(n = 1) %>% ungroup()
  rho_ha <- cor(sampled_rows$error_f0, sampled_rows$error_fa)
  return(rho_ha)
}
sample_correlation_hh <- function(df) {
  model <- lmer(error_f0 ~ 1 + (1 | Task), data = df)
  s <- icc(model, ci = TRUE)
  est <- c(s$ICC_adjusted[2], s$ICC_adjusted[3])
  return(est)
}
sample_ratio <- function(df) {
  sampled_rows <- df %>% group_by(Task) %>% slice_sample(n = 1) %>% ungroup()
  var_f0 <- mean(sampled_rows$error_f0^2, na.rm = TRUE)
  var_fa <- mean(sampled_rows$error_fa^2, na.rm = TRUE)
  est <- sqrt(var_fa / var_f0)
  return(est)
}
compute_sse <- function(beta_val, df) {
  with(df, {
    w_star <- w_est / beta_val
    y_hat <- w_star * fa + (1 - w_star) * f0
    sum((theta - y_hat)^2)
  })
}
sample_aversion <- function(df) {
  sampled_rows <- df %>% group_by(Task) %>% slice_sample(n = 1) %>% ungroup()
  result <- optimize(
    f = function(beta_val) compute_sse(beta_val, sampled_rows),
    interval = c(1e-6, 1.5)
  )
  return(result$minimum)
}

```

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}

for (i in 1:tasks) {
  holdout_data <- ages_responses %>% filter(Task != i)
  correlation_ha <- replicate(n_reps, sample_correlation_ha(holdout_data))
  rho_ha_range <- quantile(correlation_ha, probs = c(0.025, 0.975))
  correlation_hh <- sample_correlation_hh(holdout_data)
  rho_hh_range <- c(correlation_hh[1], correlation_hh[2])
  phi0_values <- replicate(n_reps, sample_ratio(holdout_data))
  phi_0_range <- quantile(phi0_values, probs = c(0.025, 0.975))
  beta_values <- replicate(n_reps, sample_aversion(holdout_data))
  beta_range <- quantile(beta_values, probs = c(0.025, 0.975))
}

```

Procedure

In this experiment, participants will estimate the age (in years) of an anonymous person based on a photograph. For each photograph, we will first ask the participants to enter their initial estimate. Subsequently, the next screen will show the participant an estimate generated by the AI advisor, along with a paragraph of text containing the AI's explanation for its estimate. After receiving the AI advice and reading its explanation, the participant will be asked to provide a final estimate of the age of the person in the photograph. The bonus earned by a participant will depend on the accuracy of both of their estimates – without and with AI advice.

We generated the AI advice and the accompanying explanation using OpenAI's GPT 4 tool. In an earlier study, we had observed an algorithm aversion factor β of around 0.65 in the participant responses. In this study, we specifically want to generate an environment where the advice is less accurate, and therefore the average weight-on-advice used by the participants would be closer to optimal. To achieve this, we added some random noise to the AI advice to reduce its overall prediction accuracy and simulate an environment where algorithm aversion factor is close to 1.

Prior to the main estimation task, there will be two examples to check that the participant understands the bonus payment formula. Next, the participants will be led through a set of three example questions to help them understand the task: how we are eliciting estimates and how their responses are incentivized so that it is in their interest to provide as accurate an estimate as possible on both instances (before and after AI advice). The three practice questions will be drawn from a different set of photographs which are not used in the main study.

The bonus earned from two estimates on each question are calculated independently of each other. Participants will receive a fixed payment of \$1 plus an additional bonus payment of between \$0 and \$0.10 for each estimate they provide. For every year that the estimate is off by, 1 cent will

be deducted from the maximum bonus amount of \$0.10. If this formula gives a negative value, zero bonus will be paid for that question.

Every participant will be providing two estimates for each of 10 photographs (excluding the three photographs used in the example questions). This means a total of 20 estimates from each participant, with a corresponding total bonus earning potential of \$2.

At the end of the experiment, participants will answer a short, optional demographic questionnaire asking about their age and gender.

Planned Sample Size

We plan to collect data from 1050 participants, who will each receive a random subset of 10 photographs that are evenly drawn from a data set of 200 total photographs. We expect this to yield approximately 50 pairs of participant judgments for each of the 200 photographs after exclusions (see below).

Exclusion Criteria

Participants' data will only be excluded if they:

- Complete the survey in an implausibly short amount of time (less than 3 minutes total).
- Fail to complete all the required questions on the survey (all questions except for the optional demographic questions at the end).

Hypotheses

Based on the responses received for each question, we will first run the parametric robust pivoting process described above, and then calculate our model's estimate of $\gamma_R(A)$. We will use $\gamma_R(A)$ to calculate the aggregate estimate given by $\hat{\theta} = \bar{f}^1 + \gamma_R(A)(\bar{f}^0 - \bar{f}^1)$.

Our hypotheses (corresponding to the research questions on page 1) are that:

H1. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the average initial human forecast.*

H2. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the than the average updated human forecast.*

H3. *Yes, the final estimate of the unknown quantity derived using our procedure after observing participants' judgment aggregation behavior will be more accurate than the AI advice.*

Planned Hypothesis Tests

To compare the accuracy across the different judgment aggregation methods, we will first compute the forecast errors of each method over a set of 200 items (i.e., 200 different photographs of people's faces), where the answer for question i (the true age of the person in the photograph) is denoted by θ_i .

1. The calculations described in equations (5) and (6) above will be used to calculate the forecast $\hat{\theta}$ of our proposed judgment aggregation method. The forecast error of our method will be calculated as:

$$\hat{e} = \theta - \hat{\theta}.$$

2. The forecast error of the average initial human forecast will be calculated as follows:

$$e^0 = \theta - \bar{f}^0 = \theta - \frac{1}{J} \sum_{j=1}^J f_j^0.$$

3. The forecast error of the average initial human forecast will be calculated as follows:

$$e^1 = \theta - \bar{f}^1 = \theta - \frac{1}{J} \sum_{j=1}^J f_j^1.$$

4. The forecast error of the AI advice will be calculated as follows:

$$e^a = \theta - f^a.$$

We will compare the pairwise differences in absolute errors across the 200 photographs to test hypotheses H1, H2, H3 respectively. We will use two-sided paired t-tests to make each comparison and will set $\alpha = 0.05$ as our threshold for determining statistical significance for each test.

Images From the Experimental Interface

The following screen shows the initial judgment elicitation step. In this step, the only stimulus provided is a photograph of the person whose age is to be estimated.



What is your best guess of the **age (in years)** of the person in the above photograph?

Type your answer into the text box below.



The following screen shows the updated judgment elicitation step after receiving the AI advice. In this step, in addition to the photograph of the same person, the AI advice is displayed along with a two- or three-sentence-long paragraph of text which explains the AI reasoning underlying its estimate.



Our AI system estimated that the person in the photograph is **53 years old**, offering the following explanation for its prediction:

The age estimation is based on the combination of his hair, which shows signs of thinning and receding, and his skin, which displays some wrinkles and reduced elasticity. The facial features still retain a certain level of youthfulness, yet show the beginning signs of middle age, which supports this age bracket estimation.

In consideration of this advice, what is your best guess of the age of the person in the above photograph?

Type your answer into the text box below.

